201536: metamonzogranite, Frogmore Bore

(Tuckanarra Suite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Bungar (2539)
MGA Zone 50, 626733E 6776667N

Sampled on 9 March 2017

This sample was collected from outcrop on the southwestern side of a track on Narndee Station, about 7.5 km south of Dixon Bore, 6.7 km due east of the Paynes Find – Sandstone Road, and 5.6 km north of Frogmore Bore.

Geological context

The unit sampled is a metamonzogranite dyke of the 2697–2657 Ma Tuckanarra Suite, which consists mainly of porphyritic or equigranular metamonzogranite to metagranodiorite (Van Kranendonk et al., 2013; Zibra, 2019). Plutons range from subcircular and non-deformed to elongate and strongly deformed adjacent to major shear zones (Zibra et al., 2014). Granitic rocks of the Tuckanarra Suite crosscut the 2825–2733 Ma Annean Supersuite and 2732–2686 Ma Big Bell Suite, and are intruded by the 2665–2640 Ma Jungar and 2650–2600 Ma Walganna Suites (Zibra, 2019). Tuckanarra Suite plutons also intrude supracrustal rocks of the Norie and Polelle Groups (Zibra, 2019). The metamonzogranite dyke at this locality exhibits ductile features and appears to be syn-magmatic with respect to the fine- to medium-grained seriate to equigranular biotite metamonzogranite into which it is intruded. A gneissic monzogranite of the Tuckanarra Suite collected about 18.2 km to the west-northwest yielded an igneous crystallization age of 2679 ± 8 Ma (GSWA 185997, Wingate et al., 2015a). A porphyritic monzogranite of the Tuckanarra Suite collected about 17.7 km to the west-southwest yielded an igneous crystallization age of 2681 ± 5 Ma (GSWA 198212, Wingate et al., 2015b).

Petrographic description

The sample is a fine- to medium-grained metamonzogranite, consisting of about 35% plagioclase, 30% K-feldspar, 25–30% quartz, 3% biotite–chlorite, 2% muscovite, 1–2% magnetite–hematite, 1% myrmekite, and trace apatite and zircon. Plagioclase is anhedral to subhedral, mainly equant, and up to 1.5 mm in size, although there are also some tabular phenocrysts up to 4 mm long. Twinning is poorly developed and obscured in some crystals by alteration to sericite and muscovite. K-feldspar is anhedral, up to 2 mm in size, and exhibits fine microcline twinning. Quartz forms anhedral, moderately strained grains up to 2 mm in size, as well as aggregates up to 3.5 mm in diameter composed of grains with sutured, intergrown boundaries. Oval and irregular lobes of myrmekite at the edges of K-feldspar are up to 0.5 mm in size and have smoothly curved margins. Biotite, partially replaced by chlorite, forms anhedral to subhedral laths up to 1 mm long, many of which are crudely aligned in short folia that define an incipient foliation. Muscovite is subhedral, non-aligned, up to 0.7 mm in size, and has partly replaced plagioclase. Magnetite is subhedral to euhedral, equant, and up to 0.35 mm in diameter, and partially replaced by hematite. Bladed euhedral apatite is up to 0.1 mm long. Apart from the crude alignment of biotite folia, the sample lacks a penetrative foliation. However, the strain in quartz and the development of microcline and trace myrmekite are consistent with solid-state deformation at low to moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, mainly subhedral to euhedral, and slightly rounded. The crystals are up to 350 µm long, and mainly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, most crystals are dominated by high-U, metamict domains, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 18–19 March 2021, using SHRIMP-B. Ten analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.61% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.25%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-one analyses were obtained from 20 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Figure 1. Hand specimen image of sample 201536: metamonzogranite, Frogmore Bore

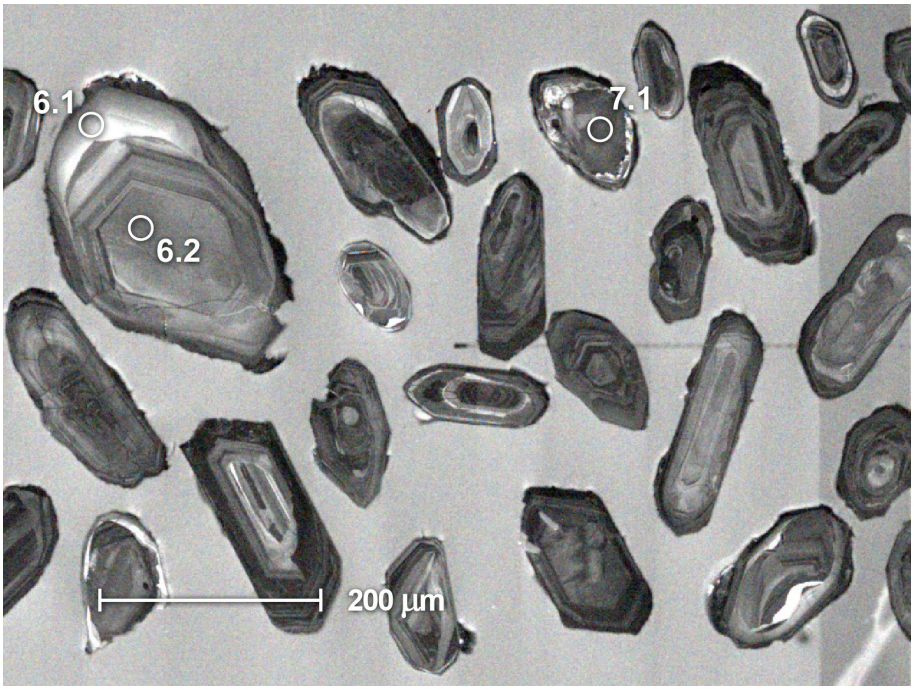

Figure 2. Cathodoluminescence image of representative zircons from sample 201536: metamonzogranite, Frogmore Bore. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 201536: metamonzogranite, Frogmore Bore

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	4	4.1	55	34	0.62	0.261	1.949	0.052	0.18379	0.00149	1.954	0.052	0.18145	0.00168	2664	58	2666	15	0.1
I	6	6.1	33	14	0.44	0.155	1.904	0.034	0.18379	0.00323	1.907	0.034	0.18241	0.00334	2718	40	2675	30	-1.6
I	2	2.1	96	94	0.98	0.068	1.977	0.023	0.18316	0.00118	1.978	0.023	0.18256	0.00122	2637	26	2676	11	1.5
I	14	13.1	169	33	0.19	0.107	1.921	0.019	0.18466	0.00092	1.923	0.019	0.18371	0.00096	2699	22	2687	9	-0.5
I	1	1.1	133	79	0.59	0.124	1.974	0.021	0.18483	0.00103	1.976	0.021	0.18372	0.00109	2640	23	2687	10	1.8
I	5	5.1	138	40	0.29	0.081	1.957	0.021	0.18506	0.00099	1.959	0.021	0.18433	0.00103	2659	23	2692	9	1.2
I	3	3.1	76	36	0.48	0.103	1.942	0.024	0.18537	0.00232	1.944	0.024	0.18445	0.00236	2675	28	2693	21	0.7
I	18	17.1	79	33	0.42	0.124	1.976	0.025	0.18567	0.00128	1.978	0.025	0.18457	0.00136	2638	27	2694	12	2.1
I	20	19.1	77	42	0.55	0.373	2.012	0.025	0.18789	0.00127	2.019	0.025	0.18457	0.00149	2593	27	2694	13	3.8
I	13	12.1	94	71	0.76	0.071	1.999	0.035	0.18567	0.00120	2.001	0.035	0.18505	0.00124	2613	38	2699	11	3.2
I	7	6.2	209	160	0.77	0.215	1.932	0.019	0.18702	0.00134	1.936	0.019	0.18511	0.00139	2684	22	2699	12	0.6
I	17	16.1	134	109	0.82	0.036	1.950	0.030	0.18613	0.00102	1.950	0.030	0.18580	0.00103	2668	34	2705	9	1.4
I	16	15.1	93	53	0.57	-0.072	1.912	0.023	0.18554	0.00121	1.911	0.023	0.18618	0.00125	2713	27	2709	11	-0.2
I	10	9.1	179	115	0.64	0.265	1.960	0.019	0.18868	0.00091	1.965	0.019	0.18632	0.00101	2652	22	2710	9	2.1
I	11	10.1	107	109	1.01	0.045	1.936	0.022	0.18748	0.00174	1.937	0.022	0.18708	0.00176	2683	25	2717	15	1.2
I	19	18.1	82	55	0.67	-0.021	1.989	0.024	0.18707	0.00131	1.989	0.024	0.18726	0.00132	2626	27	2718	12	3.4
X	12	11.1	62	60	0.97	0.000	1.937	0.026	0.18961	0.00162	1.937	0.026	0.18961	0.00162	2683	30	2739	14	2.0
X	8	7.1	264	190	0.72	0.157	1.892	0.017	0.19813	0.00082	1.895	0.017	0.19674	0.00087	2731	21	2799	7	2.4
X	21	20.1	166	130	0.78	0.078	1.873	0.018	0.19837	0.00092	1.874	0.018	0.19768	0.00094	2757	23	2807	8	1.8
D	9	8.1	22	14	0.61	2.994	3.584	0.073	0.21337	0.00328	3.695	0.078	0.18673	0.00647	1544	29	2714	57	43.1
D	15	14.1	107	43	0.40	0.739	1.867	0.021	0.21859	0.00123	1.880	0.021	0.21203	0.00160	2749	26	2921	12	5.9

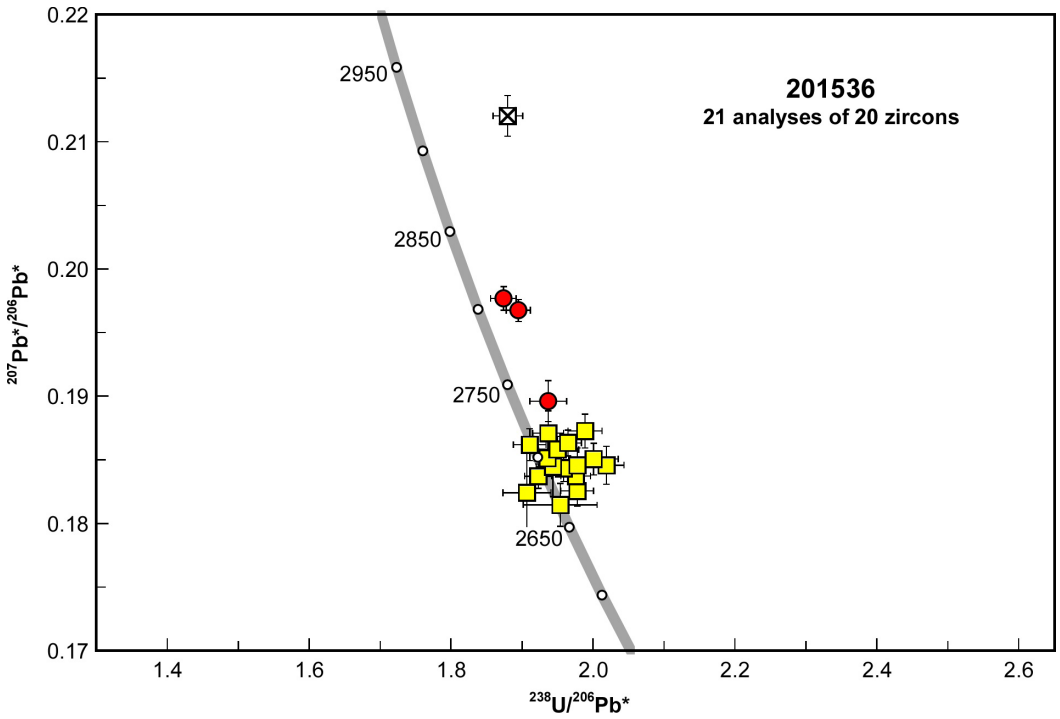

Figure 3. U–Pb analytical data for sample 201536: metamonzogranite, Frogmore Bore. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (inherited zircon cores); crossed squares indicate Group D (discordance >5%). One analysis in Group D is not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 19 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their positions within the crystals.

Group I comprises 16 analyses of 15 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2697 ± 7 Ma (MSWD = 1.3).

Group X comprises three analyses of three zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2807–2739 Ma.

The date of 2697 ± 7 Ma for the 16 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. The dates of 2807–2739 Ma for the three analyses in Group X are interpreted as the ages of inherited zircons.

References

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Recommended reference for this publication

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